

Database query examples

Most databases will use a software tool to handle the administration of a database. Most will have a graphical user interface (GUI). The one that was used in the video example is called [phpMyAdmin](#). It is one of the most popular administration tools used for MySQL and MariaDB database servers.

The 'employee' database example:

The screenshot shows the phpMyAdmin interface for a database named 'employee_test_db-32313265b2'. The left sidebar shows a tree view of the database contents, including tables like 'departments', 'dept_emp', 'dept_manager', 'employees', 'salaries', and 'titles'. The main panel displays the 'Structure' tab for the 'employees' table. It shows a list of tables with their respective actions (Browse, Structure, Search, Insert, Empty, Drop) and summary statistics (Rows, Type, Collation, Size, Overhead).

Table	Action	Rows	Type	Collation	Size	Overhead
departments	Browse Structure Search Insert Empty Drop	9	InnoDB	latin1_swedish_ci	16.0 KiB	-
dept_emp	Browse Structure Search Insert Empty Drop	~331,103	InnoDB	latin1_swedish_ci	19.0 MiB	-
dept_manager	Browse Structure Search Insert Empty Drop	24	InnoDB	latin1_swedish_ci	32.0 KiB	-
employees	Browse Structure Search Insert Empty Drop	~299,468	InnoDB	latin1_swedish_ci	14.5 MiB	-
salaries	Browse Structure Search Insert Empty Drop	~2,838,426	InnoDB	latin1_swedish_ci	95.6 MiB	-
titles	Browse Structure Search Insert Empty Drop	~441,786	InnoDB	latin1_swedish_ci	18.6 MiB	-
6 tables	Sum	~3,910,816	InnoDB	latin1_swedish_ci	147.8 MiB	0 B

Example of a table structure:

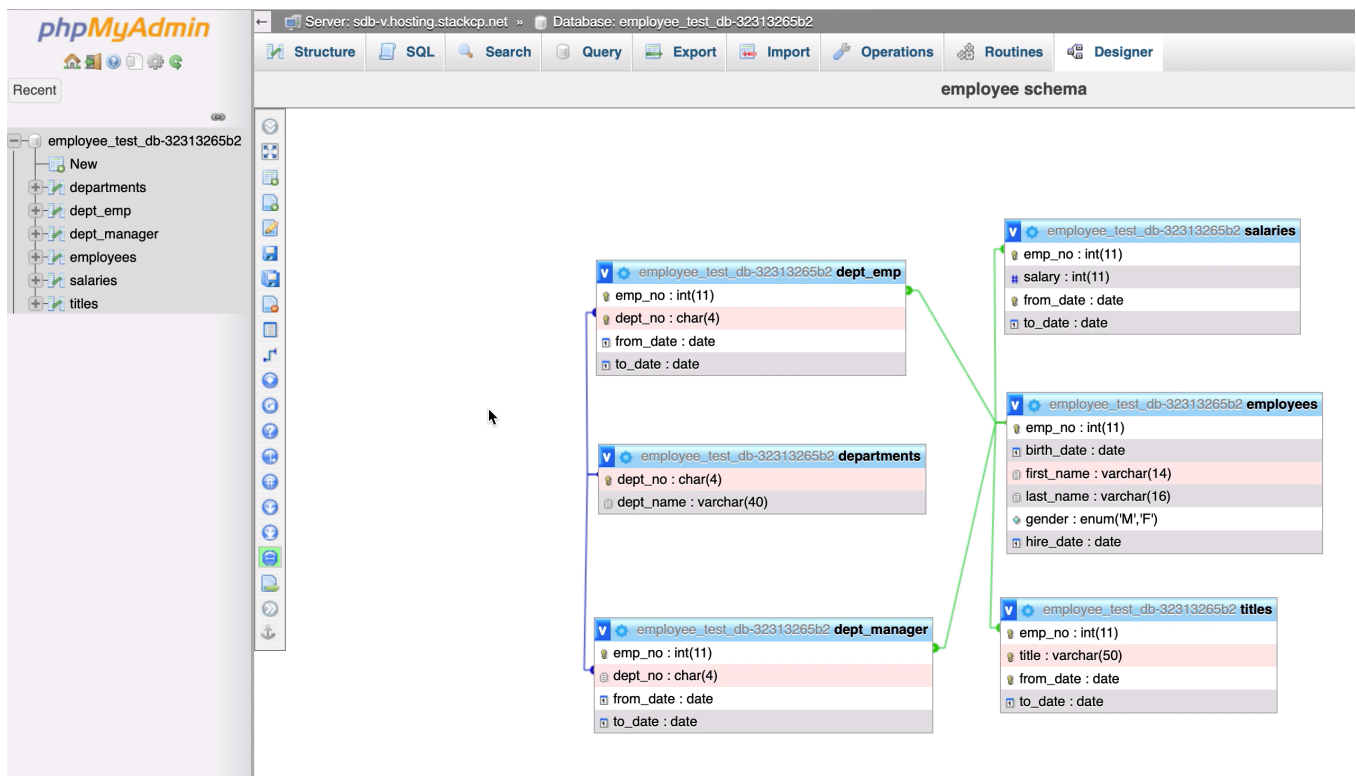
The screenshot shows the 'Structure' tab for the 'employees' table in phpMyAdmin. It displays the table structure with columns, their types, collations, attributes, nullability, defaults, and comments. Below the table structure, there are options to add new columns, indexes, and other table-related actions.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	emp_no	int(11)			No	None			Change Drop More
2	birth_date	date			No	None			Change Drop More
3	first_name	varchar(14)	latin1_swedish_ci		No	None			Change Drop More
4	last_name	varchar(16)	latin1_swedish_ci		No	None			Change Drop More
5	gender	enum('M', 'F')	latin1_swedish_ci		No	None			Change Drop More
6	hire_date	date			No	None			Change Drop More

Below the table structure, there are options to add new columns, indexes, and other table-related actions. The 'Indexes' section shows the primary key for the 'emp_no' column.

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	emp_no	299468	A	No	

Example of a Database schema:



It is a useful interface for beginners to writing queries as you can also use its graphical interface to create a query instead of directly writing SQL code.

Once you have setup your database, you can retrieve, update and maintain those records. You can also run queries on that data.

A query is a request for information from a database. You can use queries to retrieve data and add or remove data. Most queries will return data from multiple tables. There are many types of queries that can be used, and some are used by businesses to generate reports.

Common database queries include:

- **Select** – this type of query retrieves (reads) data and allows you to view it
- **Action** – this type of query allows you to perform a task relating to the data – common types include update, append, make-table, and delete

You would make a query from using a query language - SQL (Structured Query Language)

Different types of queries

SELECT: This allows you to see a set of data within the database.

In the following example we are asking for those employees who's job title matches 'Senior Engineer':

```
SELECT
    `employees`.*
FROM
    `titles`
LEFT JOIN `employees` ON `titles`.`emp_no` = `employees`.`emp_no`
WHERE
    `titles`.`title` = 'Senior Engineer';
```

Server: sdb-v.hosting.stackcp.net » Database: employee_test_db-32313265b2 » Table: employees

Showing rows 0 - 24 (97750 total, Query took 0.0006 seconds.)

SELECT `employees`.* FROM `titles` LEFT JOIN `employees` ON `titles`.`emp\_no` = `employees`.`emp\_no` WHERE `titles`.`title` = 'Senior Engineer';

Options

emp_no	birth_date	first_name	last_name	gender	hire_date
10001	1953-09-02	Georgi	Facello	M	1986-06-26
10003	1959-12-03	Parto	Bamford	M	1986-08-28
10004	1954-05-01	Chirstian	Koblick	M	1986-12-01
10006	1953-04-20	Anneke	Preusig	F	1989-06-02
10009	1952-04-19	Sumant	Peac	F	1985-02-18
10012	1960-10-04	Patricio	Bridgland	M	1992-12-18
10018	1954-06-19	Kazuhide	Peha	F	1987-04-03
10026	1953-04-03	Yongqiao	Berztiiss	M	1995-03-20
10027	1962-07-10	Divier	Reistad	F	1989-07-07
10029	1956-12-13	Otmar	Herbst	M	1985-11-20
10030	1958-07-14	Elvis	Demeyer	M	1994-02-17
10031	1959-01-27	Karsten	Joslin	M	1991-09-01
10032	1960-08-09	Jeong	Reistad	F	1990-06-20
10035	1953-02-08	Alain	Chappelet	M	1988-09-05
10037	1963-07-22	Pradeep	Makrucki	M	1990-12-05
10040	1959-09-13	Weiyi	Meriste	F	1993-02-14
10043	1960-09-19	Yishay	Tzvieli	M	1990-10-20
10047	1952-06-29	Zvonko	Nyanchama	M	1989-03-31
10051	1953-07-28	Hidefumi	Caine	M	1992-10-15
10056	1961-09-01	Brendon	Bernini	F	1990-02-01
10057	1954-05-30	Ebbe	Callaway	F	1992-01-15
10062	1961-11-02	Anoosh	Peyn	M	1991-08-30
10063	1952-08-06	Gino	Leonhardt	F	1989-04-08
10066	1952-11-13	Kwee	Schusler	M	1986-02-26
10067	1953-01-07	Claudi	Stavenow	M	1987-03-04

UPDATE: You can use this to update a set of data in the database.

In our example we updated a specific record job title from 'staff' to 'Engineer':

```
UPDATE
  `titles`
SET
  `title` = 'Engineer'
WHERE
  `emp_no` = '10002';
```

INSERT: This allows you to insert new data into a table within the database.

In our example we inserted a new record/row (department) into our 'departments' table.

```
INSERT INTO `departments`(`dept_no`, `dept_name`)
VALUES('d010', 'test department')
```

DELETE: This lets you delete data from a table in the database.

In our example we deleted the 'test department' (record/row) from the 'departments' table.

```
DELETE
FROM
  `departments`
WHERE
  `dept_no` = 'd010';
```

Other SQL queries

CREATE TABLE: You can use this to create a table within the database.

```
CREATE TABLE `employee_test_db-32313265b2`.`Hobby` (  
  `emp_no` INT(11) NOT NULL,  
  `hobby` CHAR(40) NOT NULL,  
  PRIMARY KEY(`emp_no`)  
) ENGINE = InnoDB;
```

ALTER TABLE: This gives you the option to alter a table in the database, such as adding a new column.

```
ALTER TABLE  
  `Hobby` ADD `hobby2` CHAR(40);
```

DROP TABLE: You can use this to delete a table from the database.

```
DROP TABLE  
  `Hobby`;
```

Using the Query builder:

You can also build queries using the [graphical user interface](#):

The screenshot displays the Query Builder interface with the following components:

- Navigation Bar:** Structure, SQL, Search, Query, Export, Import, Operations, Routines, Designer.
- Query window:** Multi-table query (selected), Query by example.
- Table Selection:** Three panels for 'departments', 'dept_manager', and 'employees'. Each panel includes a 'Show' checkbox, 'Table alias' and 'Column alias' input fields, and a '+ criteria' button.
- SQL Code:**

```

1 SELECT `departments`.*, `dept_manager`.*, `employees`.*
2 FROM `departments`
3     LEFT JOIN `dept_manager` ON `dept_manager`.`dept_no` = `departments`.`dept_no`
4     LEFT JOIN `employees` ON `dept_manager`.`emp_no` = `employees`.`emp_no`;

```
- Buttons:** Update query, Submit query.
- Execution Status:** Showing rows 0 - 23 (24 total, Query took 0.0007 seconds.)
- SQL Code (Repeated):**

```

SELECT `departments`.*, `dept_manager`.*, `employees`.*
FROM `departments`
LEFT JOIN `dept_manager` ON `dept_manager`.`dept_no` = `departments`.`dept_no`
LEFT JOIN `employees` ON `dept_manager`.`emp_no` = `employees`.`emp_no`;

```
- Profiling:** [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
- Table Controls:** Show all, Number of rows: 25, Filter rows: Search this table, Sort by key: None.
- Result Table:**

dept_no	dept_name	emp_no	dept_no	from_date	to_date	emp_no	birth_date	first_name	last_name	gender	hire_date
d001	Marketing	110022	d001	1985-01-01	1991-10-01	110022	1956-09-12	Margareta	Markovitch	M	1985-01-01
d001	Marketing	110039	d001	1991-10-01	9999-01-01	110039	1963-06-21	Vishwani	Minakawa	M	1986-04-12
d002	Finance	110085	d002	1985-01-01	1989-12-17	110085	1959-10-28	Ebru	Alpin	M	1985-01-01
d002	Finance	110114	d002	1989-12-17	9999-01-01	110114	1957-03-28	Isamu	Legleitner	F	1985-01-14
d003	Human Resources	110183	d003	1985-01-01	1992-03-21	110183	1953-06-24	Shirish	Ossenbruggen	F	1985-01-01
d003	Human Resources	110228	d003	1992-03-21	9999-01-01	110228	1958-12-02	Karsten	Sigstam	F	1985-08-04
d004	Production	110303	d004	1985-01-01	1988-09-09	110303	1956-06-08	Krassimir	Wegerle	F	1985-01-01
d004	Production	110344	d004	1988-09-09	1992-08-02	110344	1961-09-07	Rosine	Cools	F	1985-11-22
d004	Production	110386	d004	1992-08-02	1996-08-30	110386	1953-10-04	Shem	Kieras	M	1988-10-14
d004	Production	110420	d004	1996-08-30	9999-01-01	110420	1963-07-27	Oscar	Ghazalie	M	1992-02-05
d005	Development	110511	d005	1985-01-01	1992-04-25	110511	1957-07-08	DeForest	Hagimont	M	1985-01-01
d005	Development	110567	d005	1992-04-25	9999-01-01	110567	1964-04-25	Leon	DasSarma	F	1986-10-21

This is useful and a good way to learn how to build queries as it also displays the SQL code.